

essed/formatted in a high-level data link control (HDLC) frame with CRC for data validity check as well as data delimiter receiver. The formatted data is transmitted at 300 bps to the DRT transponder of the INSAT satellite by using a suitable channel-access protocol. The data is transmitted in burst

mode and repeated once for reliable data reception.

The satellite converts the UHF band into C-band and transmits to the 3.8m antenna receive ground (hub) station. The hub station receives real-time GPS data from all the landslide-prone sites and the signal is demodulated

to recover the data at 300 bps. The recovered data is fed to the data processing unit of the data processing centre. This real-time data is then transferred to computers for modeling the analysis of each site.

Thus the communication system can be divided into four parts: satellite transmit unit (UHF transmitter), space segment, receive station (hub) and channel access protocol.

Satellite transmit unit. This UHF transmitter has been designed and developed by Space Applications Centre, ISRO, for meteorological data collection application using the UHF transponder of INSAT

series satellite. The block diagram of the UHF transmitter is shown in Fig. 4. It consists of a digital message generator, BPSK modulator, UHF synthesiser, power amplifier and nearly omnidirectional patch antenna.

The digital processor receives data from the multi-antenna GPS receiver. The GPS time reference is used for transmission schedule. Hence it can also transmit in TDMA mode. The data received is formed in an HDLC-like frame and convolution coded. Then the data stream along with overhead symbols is transmitted to the satellite. The overhead preamble consists of 128 symbols of carrier recovery, 64 symbols of clock recovery and 64 symbols of unique word (UW) added at the beginning of the coded data packet.

The complete frame is given to the BPSK modulator at 600 symbols per second rate of modulation. Direct modulation is done at transmit-carrier frequency of 402.75 MHz, which is selected from the frequency synthesiser. The modulated signal is then amplified to 10W by the power amplifier and transmitted to the satellite through a patch antenna, which has nearly half hemispherical coverage.

This unit can be programmed for transmitting the same data packets more than one time. It can also be programmed for fixed or random time gap between two packets.

Burst format (or channel frame structure). The GPS multi-antenna unit receives data from different antennae in switching mode. All the antennae are given identification numbers (ANT. ID) of eight bits. After processing, the data is transferred to the satellite transmit unit. The size of the data packet for each antenna is maximum 40 characters (time, altitude, latitude and longitude). The satellite transmit unit collects (every 10 minutes) data from different antennae one by one and adds the site ID of eight bits. Then each packet of data is formatted in HDLC form, scrambled, forward-error

TABLE I
Broad Specifications of UHF Transmitter

Parameter	Value
Transmit frequency range	402.75 MHz $\pm$ 100 kHz
Output power	10W
Modulation	BPSK
Data rate	300 bps (1/2 rate)
Antenna pattern and polarisation	Nearly omni. linear/circular
Data size	As given
Time of transmission	As per transmission schedule
Forward error correction	1/2 rate convolution coding Constrain length, k=7, G1=171, G2=133
Transmission format	As given in Fig. 4
CR	128 symbols (all '1')
BTR	64 symbols, alternate '1' and '0'
UW	64 symbols (07EA CDDA 4E2F 28C2)
Error detection polynomial	16 bits, CCITT CRC 16
Dummy bits	8 bits (all zero)

TABLE II
Satellite Parameters

Parameters	INSAT 3A	KALPANA
Rx frequency range	402.75 $\pm$ 100 kHz	402.75 $\pm$ 100 kHz
Polarisation	RHCP	LHCP
Rx G/T	-19.0 dB/K	-19.0 dB/K
Downlink frequency	4504.2 MHz	4506.05 MHz
EIRP	24 dBW	21 dBW

TABLE III
Satellite Link Calculation

UPLINK		DOWN LINK	
Parameter	Value	Parameter	Value
Uplink frequency	: 402.750 MHz	Downlink frequency	: 4505 MHz
Uplink path loss	: 177 dB	Downlink path loss	: 197.0 dB
Transmit power	: 10 dBW	Satellite EIRP	: -1 dBW
Antenna gain (including cable loss)	: -3 dBi	G/T of earth station (3.8m)	: 17 dB/K
Transmit EIRP	: 7 dBW	Downlink C/N ₀	: EIRP path loss + G/T + K : 47.6 dBHz
Satellite G/T	: -19 dB/K		
Uplink C/N ₀	: EIRP-path loss + G/T + K : 39.6 dBHz		